On the Stiffness of Differential Equations

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Stiff Differential Equations.

Motivating Examples: ODE.

Let's consider the following simple ordinary differential equation (ODE):

$$y'(t) = -C(y(t) - \cos(t)) - \sin(t), \quad y(0) = y_0,$$

where C is a positive real number. Its analytical solution is given by

$$y(t) = \cos(t) + (y_0 - 1)e^{-Ct},$$

as we consider a perturbation term

$$\delta(t) = y(t) - \cos(t).$$

The term e^{-Ct} vanishes as t gets bigger. This means that the solution to the ODE is similar to $\cos(t)$ for large t. The constant $C > 0$ controls the decay rate of e^{-Ct}. Now let's fix $C = 10^2$ and assume that we do not know the analytic solution. Then we must seek a numerical solution.

The simplest method is the forward Euler method:

$$y(t_{n+1}) = y(t_n) + \Delta t \left(-10^2 y(t_n) + 10^2 \cos(t_n) - \sin(t_n)\right), \quad y(t_0) = y_0.$$

Once we fix the time step size Δt, then the recurrence relation produces a sequence of numerical solution $\{y(t_n)\}_{n=0}^{\infty}$. The time step Δt plays a crucial role for the accuracy of the numerical solution. To $\{y(t_n)\}_{n=0}^{\infty}$ converge, $\Delta t < 2/100$ is required. Otherwise, the sequence oscilates with increasing amplitude. This can be generalized to the step size requirement of $\Delta t < 2/C$. Hence, the larger coefficient, the smaller step size.

In general, the term "stiffness" refers to the phenomenon that explicit methods does not work well. In other words, explicit methods require exceedingly small step size when there is stiffness. In our example, this was due to a large coefficient $C = 10^2$. One may argue that, change of variables can absorb large coefficients. In this case, of course we can increase Δt, yet the length of the time interval should increase together in order to simulate the same dynamics, so the relative step size remains constant.

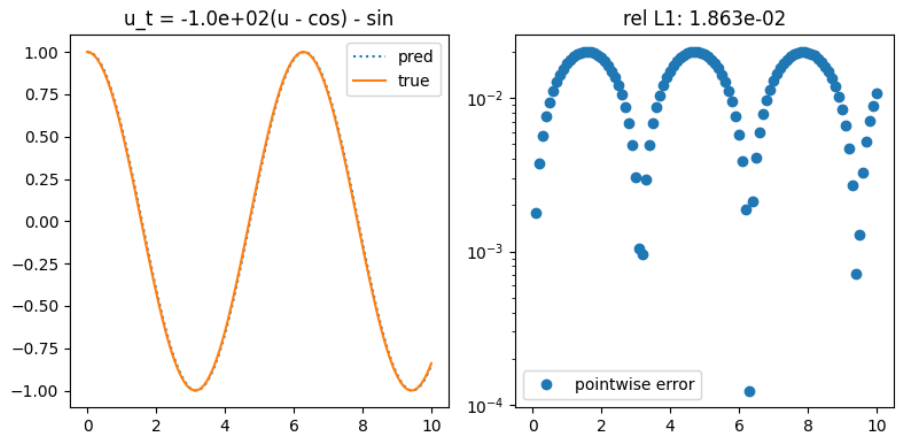

Figure 1: Forward Euler for the stiff ODE

Motivating Examples: PDE.

The stiffness originated from large coefficients of ODEs can explain stiffness in PDEs. The fundamental reason is that, numerical methods to solve PDEs usually convert PDE to a system of ODEs. To get a grasp, let us begin with a concrete example.

Let's consider the Kuramoto-Sivashinsky equation, which reads

$$\partial_t u + \partial_x^2 u + \partial_x^4 u + u\partial_x u = 0,$$

on a periodic spatial domain $(0, 32\pi)$. Due to the periodicity of the spatial domain, Fourier based methods would be the best choices. Here, we consider the Fourier spectral-collocation method, since it deals with nonlinear terms easily.

Assume that we have discretized the spatial domain by N uniform grid, $x_j = 32\pi(j-1)/N$, for $j = 1, \ldots, N$ and for some even number N. Let $P = 32\pi$ and $k \in \{-N/2+1, \ldots, N/2\}$. Then Fast Fourier Transform (FFT) computes

$$\hat{u}_k := \mathcal{F}[u](k) = \sum_j u(x_j) e^{-i\frac{2\pi k}{P}x_j}.$$

We denote $\hat{u} : k \mapsto \hat{u}_k$. The inverse transform computes

$$\tilde{u}(x_j) := \mathcal{F}^{-1}[\hat{u}](x_j) = \frac{1}{N}\sum_k \hat{u}_k e^{i\frac{2\pi k}{P}x_j}.$$

Note that $\tilde{u}(x_j) = u(x_j)$ holds exactly. Thus $\tilde{u}$ is a trigonometric interpolant of u at $\{x_j\}_{j=1}^N$. Fourier transform converts $\partial_x u$ to the multiplication of $i \cdot \xi$ and corresponding Fourier modes, where $\xi = \frac{2\pi k}{P}$ is the wavenumber. That is,

$$\mathcal{F}[\partial_x u](k) = i\xi\hat{u}_k.$$

The Fourier spectral-collocation method converts the Kuramoto-Sivashinsky equation to the following system of ODEs:

$$\partial_t \hat{u}_k(t) = -(i\xi)^2 \hat{u}_k(t) - (i\xi)^4 \hat{u}_k(t) - \frac{1}{2}(i\xi)\mathcal{F}\left[\mathcal{F}^{-1}\left[\hat{u}_k(t)\right]^2\right].$$

Stiffness occurs due to $\xi^2 - \xi^4$, which is the coefficient of the linear term $\hat{u}_k(t)$. For example, let's fix $N = 128$. Then, for $k = N/2 = 64$, $\xi^2 - \xi^4 = 4^2 - 4^4 = -240$. Thus, for the forward Euler time discretization, the step size Δt should be less than $2/240$ at least. However, the Kuramoto-Sivashinsky equation exhibits chaotic long-time behavior, which is of the most interests. If simulation time $t_{\max}$ is large, say 90, then the forward Euler method requires at least $90 \times 120 = 10800$ steps. Finer time discretizations are necessary to compute accurate solutions.